CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (E & C) Examination

Nov/Dec 2013

EC – 309 Digital Communication

Date: 04.12.13, Wednesday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks: 70**

***Instructions*:**

(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4)Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1 A An analog signal is expressd by the equation $x(t)=3 \cos 50\pi t+ 10 \sin 300\pi t-\cos 100\pi t$.Calculate the nyquist rate for this signal. **3**

B What is the necessity of non-uniform Quantization? Explain A-law and μ-Law comapanding to achieve no-uniform Quantization. **4**

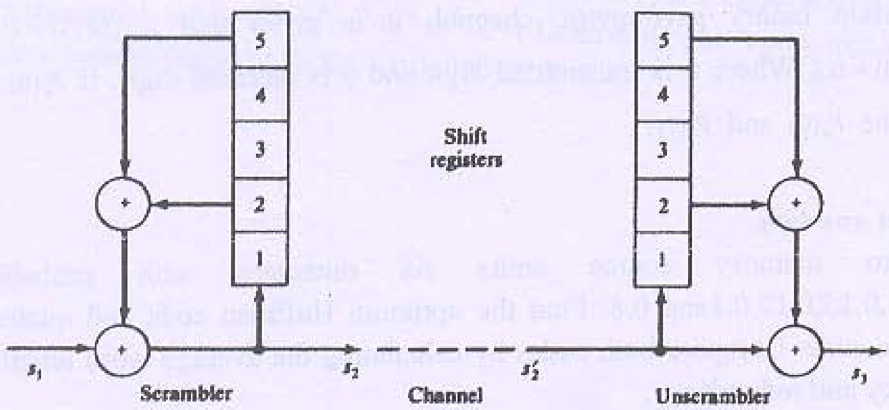
C Explain the delta modulation and its advantage over pulse-code modulation (PCM). **4**

Q-2 **Attempt any two.**

A What are the line codes? Calculate the power spectrum density of on-off signaling. **6**

B What are coherent and non-coherent digital modulation techniques? Explain the generation and demodulation of cohenrent ninary amplitude shift keying (ASK). **6**

C Illustrate the operation of data scrambler shown in figure below using data input $s_1 = 100000000000$ **6**



Q-3 **Attempt any two.**

A Explain the generation and detection of coherent binary Frequency shift keying (BFSK) modulation technique. Draw the power spectrum density and a constellation diagram for the BFSK. Also calculate its minimum bandwidth **6** requirement.

B In a digital communication system, the bit rate of the NRZ data stream is 1 MbPS and the carrier frequency is 100 MHz. Find the symbol rate of transmission and bandwidth requirement of the channel in the following cases of different techniques used. 6

(i) BPSK system

(ii) QPSK system

(iii)16-PSK system

C Explain the signaling with controlled ISI. 6

SECTION – II

Q-4 A A card is drawn randomly from a regular deck of cards. Assign probability to the event that the card drawn is (a) red card (b) a balck queen (c) a picture card (d) a number card with number 7 (e) a number card with number less than 5. 3

B A random variable X has the probability density 3

$$f_X(x) = \begin{cases} \frac{\pi}{16}\cos\frac{\pi x}{8} & -4 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

Find its mean value,second moment and its variance.

C A dice is tossed once. Let X denotes its outcome. 3

(a) Find the cumulative distribution function and draw the graph of the CDF

(b) Find the probability P(3≤ X<5)

D For certain binary asymmetric channel, it is given that $P_{y/x}(0|1) = 0.1$ and $P_{y/x}(1|0) = 0.2$,Where x is transmitted digit and y is received digit. If $P_x(0) = 0.4$, determine $P_y(0)$ and $P_y(1)$. 2

Q-5 **Attempt any two.**

A A Zero memory source emits six messages with probabilities 0.3,0.25,0.15,0.12,0.1and 0.8. Find the optimum Huffman code and quaternary Huffman code. Compare both codes by calculating the average word length ,the efficiency and redundancy. 6

B Find the channel capacity of Binary Symmetric Channel (BSC). 6

C Obtain the generator matrix for a systematic (7,4) cyclic code if G(p)=p^3 + p+1.Also obtain the parity check matrix. 6

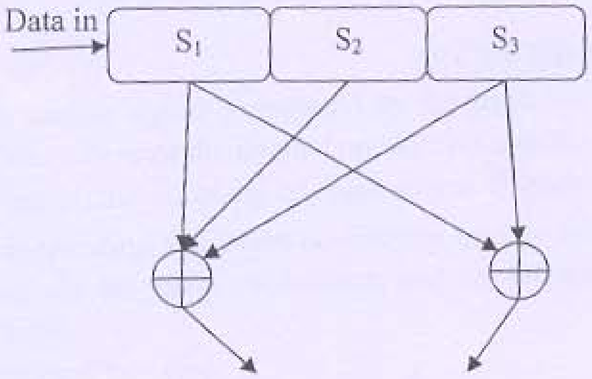
Q-6 **Attempt any two.**

A The parity check matrix of a particular (7,4) linear block code is expressed as 6

$$[H]= \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Obtain generator matrix (G)

B For the convolution encoder shown in the figure below, draw its state diagram, Tree code and Trellis diagram. 6



C (a) Show that the channel capacity of being an ideal AWGN Channel with infinite bandwidth B is given by $C_\infty = 1.44\frac{S}{\eta}$, Where noise power $N = \eta B$, S is the average signal power and $\eta/2$ is the power spectrum density of white Gaussian noise. 3

(b) Three error correcting (23,12) Golay code is a cyclic code with a generator polynomial $G(x)=x^{11}+x^9+x^7+x^6+x^5+x+1$.Determine the code words for the data vectors 000011110000,101010101010 3